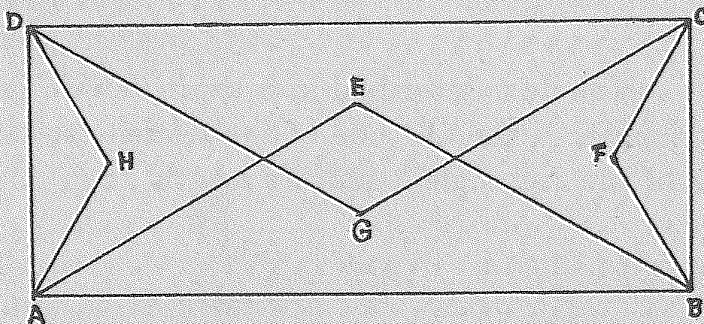


ITMA 96

SCREENING TEST - JUNIOR LEVEL - 1996

1. In the formula $C = \frac{en}{R + nr}$, e, R and r are positive constants, $n > 0$. When n increases C
- (A) decreases (B) increases (C) remains constant
(D) increases upto $n = n_0 > 0$ and then decreases
2. If $f(x) = \frac{x(x-1)}{2}$, then $f(x+2)$ equals
- (A) $(x+2)f(x)$ (B) $x(x+2)f(x)$ (C) $\frac{xf(x)}{x+2}$ (D) $\frac{(x+2)f(x+1)}{2x}$
3. If the expression $21x^2 + ax + 21$ can be factored into two linear binomial factors with integer co-efficients, a
- (A) can be any odd number (B) is some odd number
(C) can be any even number (D) is some even number
4. A three digit number N has the digits, h, t and u from left to right with $h > u$. When the number got by reversing the digits is subtracted from N , the unit digit of the difference is 4. The tens and hundreds digits of the difference are respectively
- (A) 5,9 (B) 9,5 (C) 5,4 (D) 4,5
5. The sides of a triangle are 6 units, 12 units and 12 units. The radius of the circle passing through all the three vertices of the triangle is
- (A) $\frac{7\sqrt{15}}{5}$ (B) $4\sqrt{3}$ (C) $6\sqrt{3}$ (D) $\frac{8\sqrt{15}}{5}$
6. A rectangle is inscribed in a triangle with its length on the base of length a of the triangle. If the altitude of the triangle to this base is of length h and the breadth b of the rectangle is half its length, then
- (A) $b = \frac{1}{2}h$ (B) $b = \frac{ah}{h+b}$ (C) $b = \frac{ah}{a+2h}$ (D) $b = \sqrt{\frac{ah}{2}}$

7. P is any interior point of an equilateral triangle ABC . The perpendicular distances from P to the sides AB, BC and CA of the triangle are respectively h_1, h_2, h_3 . Then $h_1 + h_2 + h_3$ is
- (A) not a constant with reference to the triangle
 (B) greater than the altitude of the triangle
 (C) less than the altitude of the triangle
 (D) equal to the altitude of the triangle
8. Given $a = \frac{xy}{x+y}$, $b = \frac{xz}{x+z}$, $c = \frac{yz}{y+z}$ $a, b, c \neq 0$, x equals
- (A) $\frac{abc}{ab+bc+ca}$ (B) $\frac{abc}{ab+bc-ca}$ (C) $\frac{2abc}{ac+bc-ab}$ (D) $\frac{2abc}{ab+bc-ac}$
9. The ratio of the area of the square inscribed in a semicircle to that of the square inscribed in the entire circle is
- (A) 1 : 2 (B) 3 : 4 (C) 2 : 5 (D) 2 : 3
10. If $y = f(x) = \frac{x+2}{x-1}$, then $f(y)$ is equal to
- (A) $f(x)$ (B) x (C) $x f(x)$ (D) $(x-1) f(x)$
11. The ratio of the perimeter of an equilateral triangle having an altitude equal to the radius of a given circle to the perimeter of the equilateral triangle inscribed in the circle is
- (A) 2 : 3 (B) 1 : 3 (C) 1 : $\sqrt{3}$ (D) $\sqrt{3}$: 2
12. The base of a triangle is 12 units. Two lines are drawn parallel to the base terminating on the other two sides and dividing the triangle into three equal areas. Then the length in units of the parallel lines are
- (A) $4\sqrt{3}, 4\sqrt{6}$ (B) 4, 8 (C) 6, 8 (D) 4, 6
13. In the figure below $ABCD$ is a rectangle. $(AE, AH)(BE, BF)(CF, CG)(DG, DH)$ are the pairs of trisectors of angles A, B, C, D respectively. Then the quadrilateral with H, E, F, G as vertices is



- (A) a square (B) a rectangle (C) cyclic quadrilateral (D) rhombus

14. With the use of only three different weights 1 kg, 3 kg and 9 kg, the total number of objects of different weights that could be weighed if the objects as well as the weights are allowed to be kept in either of the pans of a balance is
 (A) 7 (B) 9 (C) 11 (D) 13
15. A can run around a circular track in 40 seconds. B running in the opposite direction meets A every 15 seconds. The time B takes to run around the complete track is
 (A) 12 secs (B) 20 secs (C) 24 secs (D) 25 secs
16. For a given value of k , the product of the roots of $x^2 - 3kx + 2k^2 - 1 = 0$ is 7. The roots are
 (A) positive integers (B) negative integers
 (C) rational but not integral (D) irrational
17. m and n are any two odd numbers such that $m > n$. The largest integer which divides all possible numbers of the form $m^2 - n^2$ is
 (A) 2 (B) 4 (C) 6 (D) 8
18. S is the sum of the interior angles of a polygon P for which each interior angle is $7\frac{1}{2}$ times the exterior angle at the same vertex. Then
 (A) $S = 3060^\circ$, P may be regular
 (B) $S = 3060^\circ$, P is not regular
 (C) $S = 2700^\circ$, and P is regular
 (D) $S = 2700^\circ$, and P may or may not be regular
19. In the finite sequence of numbers of the form $P + n$, where $P = 2 \cdot 3 \cdot 5 \cdots p$, the product of all prime numbers upto the prime p and n takes successively the values $2, 3, 4 \cdots p - 1$, the number of primes is
 (A) 0 (B) p (C) n (D) $p + n$
20. If $x^2 + 2x + 5$ is a factor of $x^4 + px^2 + q$, then the values of p and q must be respectively
 (A) $-2, 5$ (B) $2, -5$ (C) $6, 25$ (D) $4, 25$
21. Two swimmers on opposite banks of a 180m broad swimming pool start to swim from one of its banks to the other by the shortest route at the rate of 6m per sec and 4m per sec. They swim back and forth for 15 mts. The number of times they pass each other, if they start simultaneously, is
 (A) 25 (B) 20 (C) 24 (D) 18

22. A student on vacation for d days observed that
- it rained 7 times morning or afternoon;
 - when it rained in the afternoon it was clear in the morning;
 - there were 5 clear afternoons;
 - there were 6 clear mornings. Then d equals
- (A) 7 (B) 9 (C) 10 (D) 11
23. Given $\frac{1+x}{1-x} = \frac{1+N}{N}$ $N > 0$, x
- can be any positive number less than 1
 - is a number less than 1
 - is zero
 - can be any non negative number
24. In the sequence $\{1, 8, 9, 64, 25, \dots\}$ the two terms which follow 25 are
- (A) 512, 49 (B) 256, 81 (C) 216, 49 (D) 216, 81
25. A quadrilateral is inscribed in a circle. If angles are inscribed in the four arcs cut off by the sides of the quadrilateral without intersecting the sides between the vertices, the sum of these angles is at least
- (A) 180° (B) 360° (C) 540° (D) 720°
26. The straight line segment AB is divided at C so that $AC = 3CB$. Circles are drawn on AC and CB as diameters and a common tangent to the circles meets AB produced at D . Then BD equals
- diameter of the smaller circle
 - radius of the smaller circle
 - radius of the larger circle
 - difference in the diameters
27. The area of a circle inscribed in an equilateral triangle is 48π sq. units. The area of the triangle in square units is
- (A) $144\sqrt{3}$ (B) $72\sqrt{3}$ (C) $48\sqrt{3}$ (D) $96\sqrt{3}$
28. $\left(\frac{2+\sqrt{3}}{\sqrt{2}+\sqrt{2+\sqrt{3}}} + \frac{2-\sqrt{3}}{\sqrt{2}-\sqrt{2-\sqrt{3}}} \right)^2$ is
- (A) 1 (B) 0 (C) 2 (D) 3
29. $\frac{444445 \times 888885 \times 444442 + 444438}{444444^2}$ is
- (A) 888 883 (B) 888 881 (C) 444 443 (D) 444 441
30. If $a + b + c = 0$, $a^2 + b^2 + c^2 = 1$, value of $a^4 + b^4 + c^4$ is
- (A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) 2